Techniques for Reducing Cache Misses

CSE Sem-4

Cache Replacement

- When cache is full, a line must be evicted from cache to make room for incoming data – the replacement policy determines which line is evicted.

- In direct mapped cache, there is no choice since the incoming line can only be placed in one location in the cache.

- But in fully associative and set associative mapping, lines can be replaced to make room for incoming data.

  - **First In First Out (FIFO) :** Replaces the **oldest line of cache memory.**

  - **Least Recently Used (LRU) :** Replaces a line which **has not been used for the longest period of time.**

  - **Most Recently Used (MRU) :** Replaces **most recently referred line**.

  - **Last In First Out (LIFO) :** Replaces the **last line of cache memory.**

First In First Out (FIFO)

Problem 1: Consider a fully associative, and a 4-way set associative cache memory with 16 lines. The request for memory blocks is in the following order : 0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 48, 2, 58, 32, 73, 92, 155. Find number of cache hit:

~~0~~ 92
~~255~~ 155
1
4
3
8
133
159
216
129
63
48
2
58
32
73

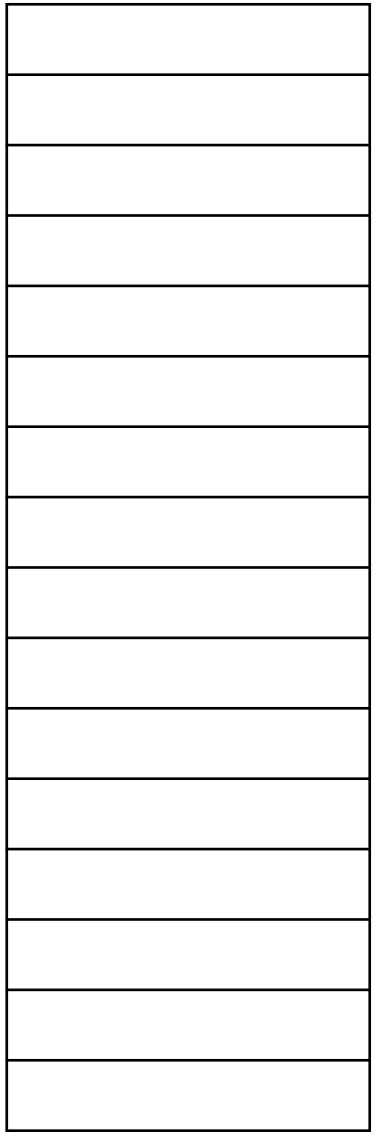
No of cache hit in Fully Associative Mapping: 0

~~0~~ 48	Set 0
~~4~~ 32	
~~8~~ 92	
216	
1	Set 1
133	
129	
73	
2	Set 2
58	
~~255~~ 155	Set 3
3	
159	
63	

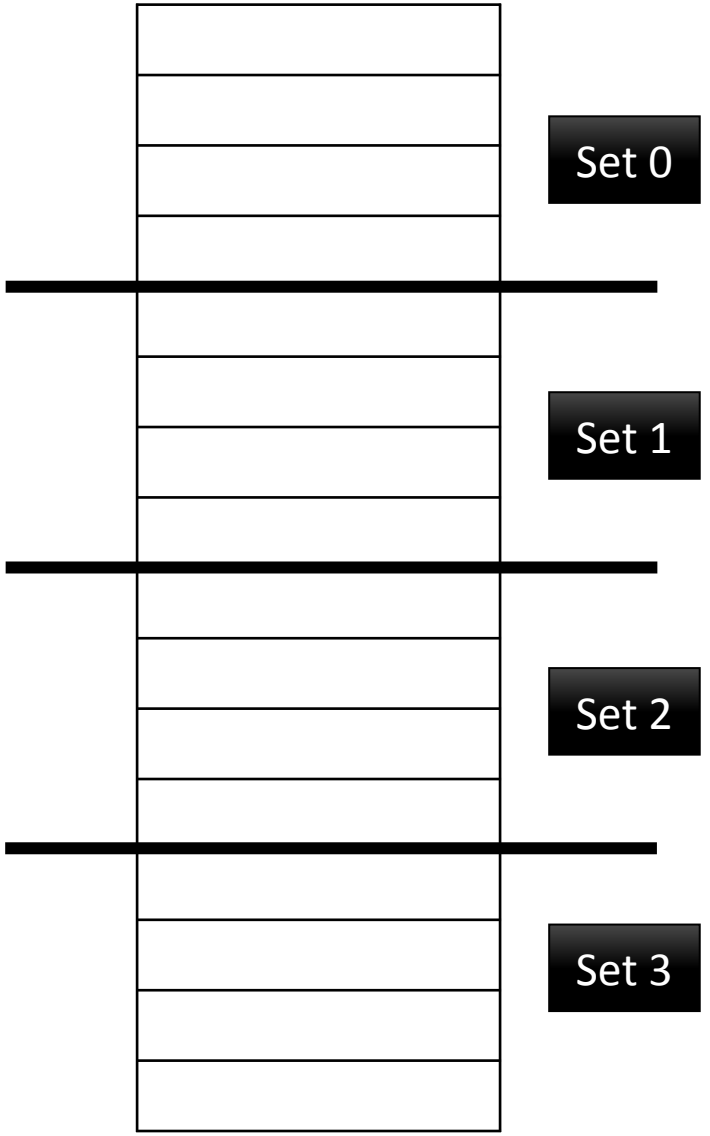
No of cache hit in 4-way set associative Mapping:

First In First Out (FIFO)

Problem 2: Consider a fully associative, and a 4-way set associative cache memory with 16 lines. The request for memory blocks is in the following order : 0, 255, 1, 4, 10, 3, 8, 133, 135, 4, 159, 255, 216, 129, 10, 63, 8, 48, 10, 133, 32, 31, 22, 0, 73, 135, 92, 159, 155, 1. Find the number of cache hits. Among 3, 133, 135 & 8 which blocks are present in the cache.



No of cache hit in Fully Associative Mapping:



No of cache hit in 4-way set associative Mapping:

Least Recently Used (LRU)

Problem 1: Consider a fully associative mapping with 8 cache lines.
The request for memory blocks is in the following order : 1, 3, 25, 8, 19, 6, 25*, 8*, 16, 35, 45, 22, 8*, 3, 16*, 25*, 7.
If LRU is used, which cache line will have memory block 7.

L0	~~1~~ 45
L1	~~3~~ 22
L2	25
L3	8
L4	~~19~~ 3
L5	~~6~~ 7
L6	16
L7	35

Least Recently Used (LRU)

Problem 2: Consider a 2-way set associative mapping with 8 cache lines.
The request for memory blocks is in the following order : 1, 3, 25, 8, 19, 6, 25*, 8*, 16, 35, 45, 22, 8*, 3, 16*, 25*, 7.
If LRU is used, which cache line will have memory block 7.

L0	8	Set 0
L1	16	
L2	~~1~~ 45	Set 1
L3	25	
L4	6	Set 2
L5	22	
L6	~~3~~ ~~35~~ 7	Set 3
L7	~~19~~ 3	

Most Recently Used (MRU)

Problem 1: Consider a fully associative mapping with 8 cache lines.
The request for memory blocks is in the following order : 1, 3, 25, 8, 19, 6, 25*, 8*, 16, 35, 45, 22, 8*, 3*, 16*, 25*,7.
If MRU is used, which cache line will have memory block 7.

L0	1
L1	3
L2	~~25~~ 7
L3	8
L4	19
L5	6
L6	16
L7	~~35~~ ~~45~~ 22

Most Recently Used (MRU)

Problem 2: Consider a 2-way set associative mapping with 8 cache lines.
The request for memory blocks is in the following order : 1, 3, 25, 8, 19, 6, 25*, 8*, 16, 35, 45, 22, 8*, 3*, 16*, 25, 7.
If MRU is used, which cache line will have memory block 7.

L0	8	Set 0
L1	16	
L2	1	Set 1
L3	~~25~~ ~~45~~ 25	
L4	6	Set 2
L5	22	
L6	~~3~~ 7	Set 3
L7	~~19~~ 35	

Last In First Out (LIFO)

Problem 1: Consider a fully associative mapping with 4 cache lines.
The request for memory blocks is in the following order : 2, 3, 4, 7, 6, 3*, 4*,7 , 5, 4*, 7, 8.
Find out the hit ratio and miss ratio.

L0	2
L1	3
L2	4
L3	~~7~~ ~~6~~ ~~7~~ ~~5~~ ~~7~~ 8

- Hit ratio: $\frac{No\ of\ hit}{Total\ Request}$ x 100%
  =3/12 =25%

- Miss ratio: $\frac{No\ of\ miss}{Total\ Request}$ x 100%
  = 9/12 =75%

Comparison Problem

Problem 1: Consider a fully associative mapping with 4 cache lines.
The request for memory blocks is in the following order : 2, 3, 4, 7, 6, 3, 4, 7, 5, 4, 7, 8.
Compare it with FIFO, LRU and MRU.

2, 3, 4, 7, 6, 3*, 4*, 7, 5, 4*, 7, 8

LIFO
2
3
4
7 6 7 5 7 8

- Hit Ratio =3/12 = 25%
- Miss Ratio = 75%

2, 3, 4, 7, 6, 3*, 4*, 7*, 5, 4*, 7*, 8

FIFO
2 6
3 5
4 8
7

- Hit Ratio = 41.67%
- Miss Ratio = 58.33%

2, 3, 4, 7, 6, 3*, 4*, 7*, 5, 4*, 7*, 8

LRU
~~2 6~~ 5
~~3~~ 8
4
7

- Hit Ratio = 41.67%
- Miss Ratio = 58.33%

2, 3, 4, 7, 6, 3*, 4*, 7, 5, 4, 7, 8

MRU
2
3
~~4 7 5 4 7~~ 8
7 6

- Hit Ratio = 16.67%
- Miss Ratio = 83.33%